

AAROHAN 2026

ASCENT OF KNOWLEDGE

Class 11th

Name

Contact Number



BITS Pilani
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 **APOGEE 2026**

Instructions:

1. This Question Paper contains 4 comprehensive passages based on the above theme. Each passage is followed by a series of questions. There will be 25 questions in all.
2. Each question has **ONLY ONE** out of the four choices as the correct answer. Marking Scheme is
 - a. **+4 for correct Answer**
 - b. **-1 for incorrect Answer**
 - c. **and 0 for not attempting the question..**
3. The questions will test your skills in Physics, Chemistry, Mathematics and Analytical Reasoning, along with your understanding of the passage. Remember, everything you need to know is already given somewhere in this passage- do not be intimidated, for winners in this game are just scared stupid explorers.

ALL THE BEST (/ ° - °) /

Space for Rough Work

ALPHYS' CATALYTIC CORE CHALLENGE

After defeating Asgore and escaping the Underground, the weight of leaving the monsters behind lingers in your heart. Drawn by guilt, you return through the molten tunnels of Hotland, where the rhythmic hum of machinery still echoes beneath the earth.

In the depths of the lab, the **CORE**—a geothermal generator that sustains the entire Underground—struggles under intense heat fluctuations. The **catalytic converter**, woven from enchanted metal and spider silk, channels raw volcanic energy into electricity. Within it lies a fragile **bio-catalyst**, an organic enzyme that accelerates the energy conversion process but risks denaturation if temperatures climb too high.

The equilibrium constant(K) is an indicator of what direction the reaction is proceeding towards. It is the ratio of the forward and the backward reaction, for $K > 1$, the reaction proceeds in the forward direction and similarly for backward reaction. At $K = 1$, the reaction is said to be at equilibrium— the reaction continues but the amount of product and reactants stay the same.

The **coolant system** meant to regulate this process is unstable, and its condition uncertain. Your task is to assess and stabilize it before the generator overheats once again.

Formulas you might require: $Q = mc\Delta T$

Specific heat capacity of the coolant is $c = 4.2 \text{ kJ kg}^{-1} \text{ K}^{-1}$. The mass of coolant in the generator is 40kg.

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1. If the heat input rate spikes to $\dot{Q}=60\text{kW}$ for 3 minutes, what is the additional temperature rise of the coolant?

- A) 7 K
- B) 64 K
- C) 68 K
- D) 15 K

2. The coolant must remain under a 15K rise per cycle. If the total heat absorbed in one cycle is $3.0 \times 10^6 \text{ J}$, which of the following is true?

- A) The coolant will exceed the safe temperature rise
- B) The coolant will remain within safe limits
- C) The coolant will exactly reach the limit
- D) Cannot be determined without knowing initial temperature

3. The decomposition in the catalytic exchanger follows $k=Ae^{-E_a/(RT)}$

If the activation energy E_a is increased, what happens to the sensitivity of the rate constant k to temperature changes?

- A) Sensitivity increases
- B) Sensitivity decreases
- C) Sensitivity remains unchanged
- D) Sensitivity becomes zero

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4. The reaction in the exchanger is endothermic. If the temperature is increased, what happens to the equilibrium constant K?

- A) K increases
- B) K decreases
- C) K remains unchanged
- D) K first increases then decreases

5. Dr. Alphys realises she has been adding a noble gas to the equilibrium inside the catalytic exchanger unknowingly. The ion exchange reaction is $A^+ + B^- \rightleftharpoons AB$, if the inert gas is added at constant pressure, what inference can be drawn to help identify problems in the generator?

- A) It shifts equilibrium to the right, speeding up the reaction.
- B) It shifts equilibrium to the left, reducing efficiency.
- C) It does not affect the equilibrium or the efficiency of the generator.
- D) Nothing can be said in this made-up game!

6. Alphys's bio-catalyst denatures above 360 K. Which graph could help Dr. Alphys best assess the rate of reaction vs. temperature for this enzyme?

- A) A steadily increasing straight line that drops at 360K.
- B) A bell-shaped curve peaking below 360 K.
- C) A bell-shaped curve peaking at 360 K.
- D) A curve that drops sharply at low temperature and peaks above 360K.

Space for Rough Work

UNDYNE'S SPEAR MECHANICS TRIAL

The fierce captain of the Royal Guard, Undyne stops you from proceeding further in your conquest to free the Underworld. It's been eons since she last saw a human, and she cannot identify what manner of being you are.

She traps you in the Aqua Arena and fights you with her many spears. The aqua arena is unique—which does not follow the rules of classical physics— gravity **flips direction**, platforms **rotate and shift**, and strange **sinusoidal energy fields** warp the potential landscape.

She starts with a rotating spear mechanism — a perfect circle of enchanted metal rotating at constant angular velocity ω . Then, the floor tiles beneath her begin to slide frictionlessly, forcing her to **conserve momentum** while she launches spears at you mid-motion. She also alters the gravitational field, the Aqua Arena begins reversing gravity for random fractions of a second, and the **effective gravity** acting on everything becomes unpredictable.

Even the **potential energy** in the field changes form, distorted by a sinusoidal pulse that turns simple oscillations into nearly chaotic motion.

Now you must solve all the following problems to survive this attack, keep in mind, with the power vested/sanctioned in you by the laws of motion and classical mechanics, you can easily solve these problems!

Space for Rough Work

7. A spear rotates in a perfect circle around a pivot with angular velocity ω . If its linear speed doubles but radius halves, what happens to angular velocity?

- A) Doubles B) Halves C) Quadruples D) Stays same

8. While sliding across the frictionless Aqua Arena, you and a platform of equal mass glide toward each other at speed v each. Just before landing on it, you hurl your sword forward (mass = $0.1m$) at speed v in the same direction of motion relative to the ground. After impact, the sword, you, and the platform all stick together and move as one. What fraction of your initial kinetic energy is lost in the process?

- A) More than 50% B) 50% C) Less than 50% D) No loss in K.E.

9. Undyne throws a spear backward while standing on a floating rock, but the rock is tethered by a loose rope to a massive wall (unmovable).

If the rope is tensionless until taut and doesn't stretch, what happens when the rope tightens?

- A) Total momentum of the *Undyne-rock-spear* system remains conserved at all times
 B) Momentum is conserved only until the rope goes taut, then external impulse breaks conservation
 C) The rope's impulse transfers momentum but not energy
 D) Momentum oscillates but total energy remains constant

10. In one sector, potential energy varies as

$$U(x) = \frac{1}{2} * k x^2 + \alpha \sin(x)$$

A small oscillation about equilibrium occurs. Which of the following is TRUE with respect to the sinusoidal term? (α is a small constant compared to k)

- A) Doesn't affect frequency
 B) Doubles effective stiffness
 C) Alters amplitude
 D) Makes motion non-periodic

Space for Rough Work

11. The Aqua Field reverses gravity for a fraction p of every second. The spear is released from a height as a free fall when the gravity just flips upward.

What can be said about average fall time for the spear from a sufficiently large height as p varies?

- A) Fall time increases linearly as p increases
- B) Fall time increases linearly as p decreases
- C) Fall time tends to infinity at $p=0.5$
- D) Fall time is the same for p and $1-p$

12. Inside Flowey's chamber, gravity reverses direction every 1.0 second, downward with acceleration $+g$ for one second, then upward with acceleration $-g$ for the next.

You release a spear from rest at height $h=4.9$ m, exactly when gravity first points downward.

Assume perfect reversals (no lag), no air resistance, and take $g=9.8$ m/s². How long will it take for the spear to reach the ground?

- A) 1s same as under constant gravity
- B) 1.4s
- C) 0.6s
- D) Never

13. If the gravity flip frequency is f flips per second, and each phase lasts $1/(2f)$ seconds, what's the average effective downward acceleration over many cycles?

- A) $g_{\text{eff}}=g$
- B) $g_{\text{eff}}=g/2f$
- C) $g_{\text{eff}}=0$
- D) None of these

Space for Rough Work

MUFFET'S SPIDER NEST CHALLENGE

Muffet, the elegant spider queen, has 50 little spiders and 50 cozy nests, numbered 1 through 50.

Each spider i is assigned to its corresponding nest i -meaning Spider 1 to Nest 1, Spider 2 to Nest 2, and so on.

But alas! Muffet is feeling playful today. When the spiders begin to crawl into their nests, Spider 1 decides to sit in a completely random nest, which may or may not be its own.

Every spider that enters afterward follows this rule:

- If their own nest is empty, they occupy it.
- If their nest is already taken, they choose another empty nest completely at random.

The spiders crawl in **sequential order** — Spider 1 first, then Spider 2, then Spider 3, and so on.

Please pay the utmost attention to the two rules. Do not misunderstand them!

Space for Rough Work

14. During the crawling process, Spider 15 enters and finds its assigned Nest 15 already occupied. Which of the following statements is NOT TRUE?

- A) Spider 1 must have taken Nest 15
- B) Spider 50 may still end up in Nest 50
- C) At this point, Nest 1 must still be vacant
- D) Spider 15's random choice may affect whether Spider 50 gets Nest 50

15. Which of the following statements is FALSE?

- A) There is no case where no spider rests in their assigned nest.
- B) The probability of Spider x ending up in Nest 1 is always less than or equal to the probability of it ending in its own nest.
- C) There is a case where all spiders, except two, occupy their assigned nests.
- D) If Nest 1 is occupied, we can guarantee that the last spider will rest in its correct nest.

16. Which nests determine the final outcome for Spider 50? (Whether they rest in their assigned nest or not)

- A) Nest 50
- B) Nest 1 and Nest 50
- C) All nests, equally distributed
- D) All nests, unequally distributed

Space for Rough Work

17. If there are only 3 spiders and 3 nests in Muffet's tea room, what is the probability that the 2nd and 3rd spiders rest in their assigned nests? (Spider 1 sits randomly)

- A) $\frac{2}{3}$, $\frac{1}{3}$
- B) $\frac{1}{3}$, $\frac{2}{3}$
- C) $\frac{2}{3}$, $\frac{1}{2}$
- D) None of the above

18. What is the probability that Spider 50 will find its assigned Nest 50 available?

- A) $\frac{49}{50}$
- B) $\frac{1}{50}$
- C) $\frac{1}{2}$
- D) None of the above

(Hint from Muffet: Try to find a relation between $P(\text{success})$ and $P(\text{failure})$. Remember, they must add up to 1, right? Also, if you're confident in your earlier answers... you're already halfway there)

19. What is the probability that Spider 49 will rest in its assigned nest?

- A) $\frac{48}{50}$
- B) $\frac{2}{50}$
- C) $\frac{2}{3}$
- D) Cannot be determined without the verdict of Spider 50.

🕸 Muffet giggles softly.

"Such delightful chaos, don't you think? Even in all that randomness humans find peace, isn't that fascinating. Anyway since you're gonna need it, here's another hint- symmetry! Like my webs!" 🕷

Space for Rough Work

MEMORY, INFORMATION & ENTROPY

Flowey, everyone's favourite evil flower is back. The root of all evil, this time has absorbed all the 6 human souls which King Asgore so dearly treasured. Flowey plans to use your soul along with the 6 souls to shatter the barrier between the Underground and the Overworld- by unleashing a quantum storm that will erase the laws of reality themselves!! Flowey has hijacked the CORE and is using its energy to create a quantum entanglement field that destabilises matter, scrambles time and unleashes total chaos- major yet a minor contribution to the universes ever increasing entropy, But as a human it is your life goal to be concerned with trivial matters such as a slight entropy spike, and prevent it, because....morals? And also because I the creator of this question paper grant you with plot armour- so now, you, last hope, must outsmart Flowey by understanding the quantum and thermodynamic tricks he's using. Only by mastering the science of uncertainty, probability, and information can you foil his plan and restore order!

Before diving in, let's take a look at these ideas:

- a) A **microstate** is a microscopic arrangement of energy of a system, it is one of the possible ways that a system's total energy can be distributed. The more microstates available, the higher the entropy.
- b) The **law of large numbers** tells us that as systems grow larger, random fluctuations average out, but the number of disordered microstates grows exponentially faster than ordered ones- making spontaneous order vanishingly rare in large systems.

Basically this means, with more data points, random fluctuations cancel out, making the average increasingly stable and predictable and individual outliers matter less as the dataset grows.

Hence, **entropy** is not only about heat or motion, but also about **uncertainty**.

Each soul that Flowey has can occupy one of the two energy levels, not knowing which soul occupies which level is the 'uncertainty' here, which will now correspond to entropy- the topic we are concerned with.

Space for Rough Work

Now Flowey **erases** (*forgets*) all records of which soul was where, and later claims that simply **remembering** it again would undo entropy—meaning entropy would be back to initial value after remembering.

In information theory, **memory** isn't some abstract thought, it's the **physical arrangement of bits, atoms, or charges that store a state**. Changing, erasing, or restoring that information **requires physical work**. According to **Landauer's principle**, every bit of information erased increases the entropy of the environment by at least

$$\Delta S = k_B \ln 2/T$$

And the total entropy is given by

$$S = -k_B \sum (x \ln(x))$$

Where S is entropy, k_B is Boltzmann's Constant = 1.38×10^{-23} Joules/ Kelvin, T is temperature in Kelvin and x is probability of a specific microstate— the summation signifies adding entropy for all microstates.

20. Flowey begins with six identical souls, all holding equal energy. A fluctuation allows each soul to occupy one of two energy levels. How many distinct microstates now exist?

- A) 6
- B) 12
- C) 64
- D) 128

21. Suddenly you notice the souls inside Flowey start vibrating faster, however total energy remains constant. Which of the following statements holds TRUE?

- A) Average potential energy drops to compensate for increase in entropy.
- B) Molecular collisions redistribute energy among more accessible states, increasing entropy.
- C) Entropy remains the same.
- D) All are false (Flowey bluffed an attack!)

22. Flowey tries to hold all energy in one configuration forever. Even if no external energy enters or leaves, the system evolves toward disorder because (consider ideal system)-

- A) Internal energy continuously converts into thermal energy, increasing total energy.
- B) Disordered configurations correspond to vastly more microstates, making them statistically dominant.
- C) The system naturally seeks to minimize total internal energy.
- D) Random molecular fluctuations bias the system toward lower entropy states only temporarily.

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23. Why is Flowey's claim about entropy being the same after remembering false?

- A) Memory has intrinsic negative entropy that cancels the disorder
- B) Recovering or erasing information requires real physical work
- C) Entropy reduction from memory recovery is offset by loss of coherence in the medium storing the data.
- D) Information retrieval only reorganizes macrostates, microscopic entropy remains unchanged.

24. If each soul could exist in one of four equally likely emotional states (joy, rage, sorrow, calm). Which of the following correctly expresses the information entropy per soul(entropy of each soul)?

- A) $S = 4 k_B \ln(0.25)$
- B) $S = 4 k_B \ln(4)$
- C) $S = k_B \ln(0.25)$
- D) $S = k_B \ln(4)$

25. If the minimum energy that must be dissipated as heat, according to Landauer's principle, to erase 1 unit of information at 300K be $x \cdot 10^{21}$ J, then x is approximately-

- A) 3
- B) 2
- C) $3 \ln(2)$
- D) $2 \ln(2)$

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ANSWER KEY

11th Grade, Undertale Theme Round 1													
1	B	2	A	3	A	4	A	5	B	6	B		
7	C	8	A	9	B	10	A	11	C	12	A	13	C
14	A	15	A	16	B	17	C	18	C	19	C		
20	C	21	B	22	B	23	B	24	D	25	A		

Solutions and hints:

ALPHYS' CATALYTIC CORE CHALLENGE

1. Heat rate* time= Heat(Q)

So, $Q(3*60) = mc\Delta T$

Difficulty: Easy

2. Use $Q = mc\Delta T$, where Q is given in J. You will see $\Delta T > 15K$.

Difficulty: Easy

3. Take natural log of the equation given and differentiate w.r.t temperature to get the equation:

$$\frac{d(\ln K)}{dT} = \frac{E_a}{RT^2}$$

If $E_a \uparrow$ the differential term increases too, thus sensitivity increases as well.

This can be done through analysis of the formula after the first differential but it's a bit vague that way.

Difficulty: Hard

4. Endothermic reactions are benefitted by an increase in temperature.

Difficulty: Easy

5. Adding an inert gas at constant pressure increases the volume of the system, lowering effective concentration of all reactants and products. Hence by Le Châtelier's principle the equilibrium shifts towards the side with more gaseous particles.

Difficulty: Easy

6. To study denaturation of the enzyme we would want a graph showing rate of reaction that peaks before 360K and then dips rapidly once denatured.

Difficulty: Easy

UNDYNE'S SPEAR MECHANICS TRIAL

7. $\omega = v/r$

Difficulty: Easy

8. Solve it yourself :) Find initial momentum, initial K.E. then find final velocity. Even if you were confused about the orientation of the sword, yourself and the platform, almost every 'valid' permutation of this system will give you a loss of $KE > 50\%$!

Difficulty: Moderate

9. When the rope gets taut(tightened), it delivers an impulsive force that abruptly stops the rock and Undyne, transferring their momentum to the wall and dissipating kinetic energy.

Difficulty: Moderate

10. Differentiate the equation w.r.t to get force, you will see the sin term shifts the equilibrium position slightly, changes the effective stiffness of the spring block system but keeps it periodic. The change in frequency is almost negligible(take limits on the sine term)

Difficulty: Hard

11. Find out effective acceleration in downward direction for $p > 0.5$, $p < 0.5$ and $p = 0.5$ and analyse.

Difficulty: Easy

12. Spear hits the ground before the gravity flips for the first time.

Difficulty: Easy

13. As long as intervals of upward and downward acceleration are the same, the effective gravity cancels out to zero

Difficulty: Easy

MUFFET'S SPIDER NEST CHALLENGE

*Analyse the true and false options in each question. It will help in solving further questions.
Now that you know all the answers- it's much easier to reason and fully understand the question.*

14. It is possible spider 1 takes nest 2 and spider 2 takes nest 15.

Option C is true because whenever any spider takes nest 1- all the spiders after that spider will simply go into their nests and the mismatch domino effect ceases.

Difficulty: Moderate

15. This was word play! There exists a case where no spider sits on- Spider 1 goes into nest 2, Spider 2 into nest 3 and so on, spider 49 in nest 50 and spider 50 in nest 1. Thus option a is false.

Option b is a very cool mathematical proof, which can be perceived logically by drawing a parallel to the principle of concentration of probability (check Monty Hall problem for a fun concentration of probability question).

Difficulty: Moderate

16. Option c of question 15 indirectly solves question 16. Consider the case where nest 1 is occupied by any spider other than the 50th, then nest 50 is necessarily empty. Thus the status Nest 1 and Nest 50 is enough information to deduce the fate for the 50th spider.

Difficulty: Hard

17. Can be solved by creating all the possible cases and then calculating the required probability.
Try it yourself now that you know the answer.

Difficulty: Hard

18. The entire question becomes a recursive problem of analysing if nest 1 or nest 50 were empty. Thus, the contribution of every other nest is already counted in the contribution of nest 1 and 50, hence calculating the probability now becomes much much simpler!

Difficulty: Hard

19. Similar to Q16 and Q18- find the nests which decide the outcome for the Spider 49, you will see only nest 1, 49 and 50 can alter the outcome for spider 49. Now you already know the denominator of your probability will be 3, now count the number of favourable outcomes.

It can be calculated mathematically with distinction that the probability that the k^{th} spider finds its nest empty is $\frac{k}{k+1}$

Difficulty: Hard

Please feel free to look into my [inspiration](#) for this question for further exposure.

[Drunk Passenger | Brainstellar Puzzles](#)

MEMORY, INFORMATION & ENTROPY

20. 6 souls, 2 possible states per soul, 2^6 possibilities.

Difficulty: Easy

21. Internal energy need not be lost to compensate for a spike in entropy. When energy is constant and there's multiple ways of arranging that energy, entropy increases.

Difficulty: Easy

22. Fundamental reason for increase in entropy, given in the paragraph.

Difficulty: Easy

23. Information is not abstract, it is stored in matter. Also given in the paragraph.

Difficulty: Easy

24. $x=0.25$, so $-k_B \sum (x^* \ln(x)) = -k_B (4 \ln(0.25)) = -4k_B \ln(0.25) = +4k_B \ln(4)$

Difficulty: Hard

25. $E_{\min} = k_B T \ln(2) = 2.87 \times 10^{-21} \approx 3 \times 10^{-21}$

Difficulty: Hard